

NAC WiFi Captive Portal System Report

Production readiness, administrator guide, and hardware integration plan for MikroTik or UniFi access points.

Prepared For

National Airports Corporation

System

NAC Smart WiFi Captive Portal

Prepared Date

20 May 2026

Current Mode

MikroTik HotSpot handoff, UniFi ready

1. Executive Summary

The NAC WiFi Captive Portal is a guest WiFi access system for visitors, passengers, staff guests, and other approved users. It provides a branded splash page, guest login, free advertisement-supported access, paid access packages, live session tracking, advertisement management, subscription plans, and controller settings for real captive portal deployment.

The system is designed so that when MikroTik or UniFi access point hardware arrives, the administrator can configure the portal URL, controller settings, SSIDs, access time, and HotSpot bridge file from the Admin Console. The goal is to make deployment practical without rewriting code.

Plain explanation: A user connects to NAC WiFi, the router or access point sends them to this portal, the user signs in or watches an advert or pays, and the router/controller then allows that device to access the internet for the approved time.

2. Report Contents

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3. System Purpose

The portal exists to control and record WiFi access. It is not just a web page. It works together with network hardware such as MikroTik RouterOS HotSpot or UniFi external portal features. The hardware blocks users before login. The portal collects the user details and chooses whether the user should receive free or paid access.

Area	Purpose
Open Portal	Public splash page for users who connect to the WiFi network.
Guest Login	Collects name, organization, phone or email before access is granted.
Free Access	Shows an active advertisement first, then creates an access grant.
Paid Access	Shows paid packages, records the payment path, and creates premium access after approval.
Admin Console	Lets staff manage sessions, ads, plans, settings, logo, and controller readiness.
Controller Integration	Connects the business flow to MikroTik or UniFi so the user actually receives internet access.

4. Basic User Flow

1. Connect

User joins the NAC WiFi SSID from phone, laptop, or tablet.

2. Redirect

MikroTik or UniFi sends the user to the captive portal splash page.

3. Register

User enters name and phone or email. The system records IP and device details where available.

4. Access

User watches an advert for free access or selects a paid package. The router/controller authorizes internet access.

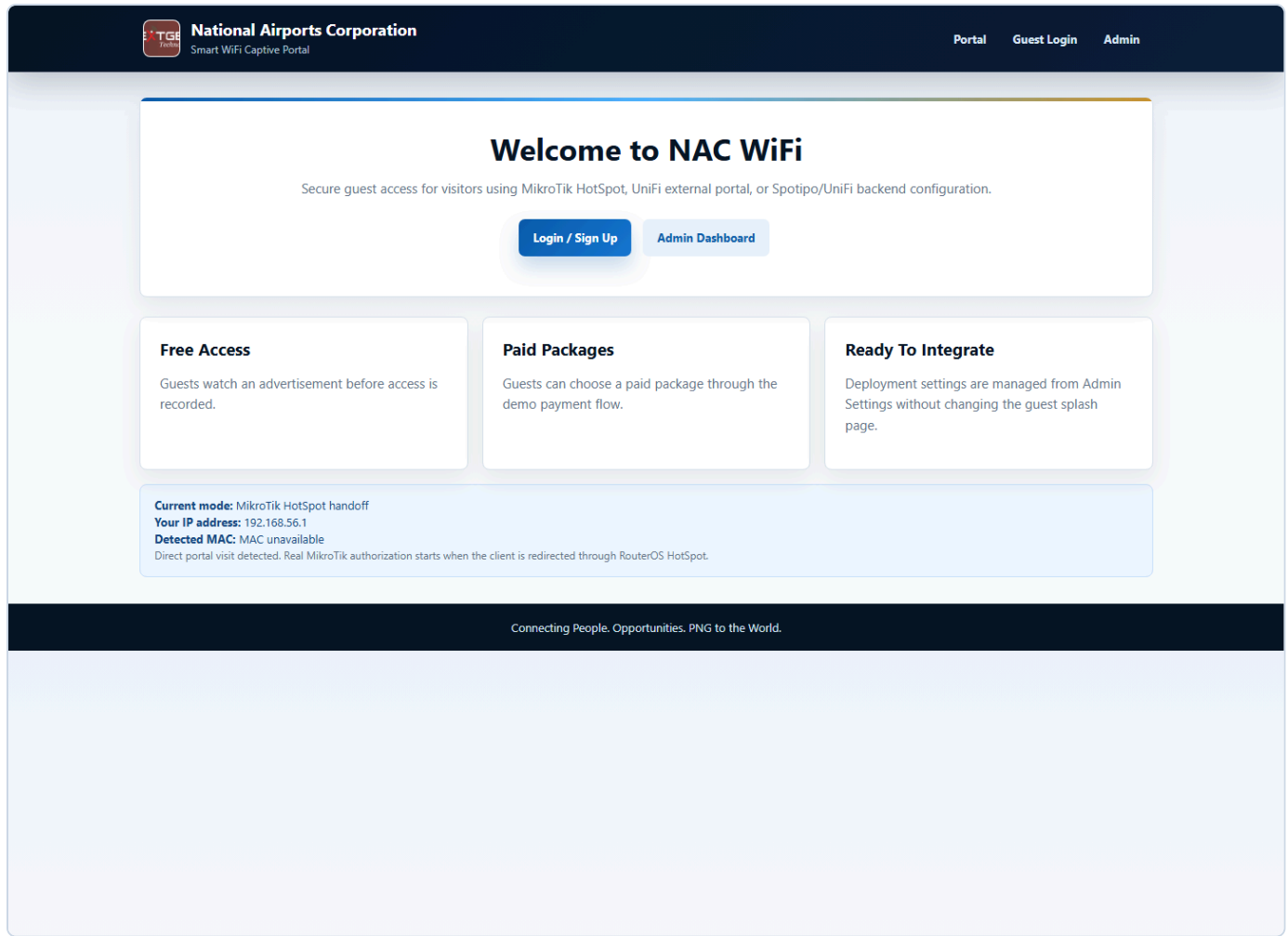


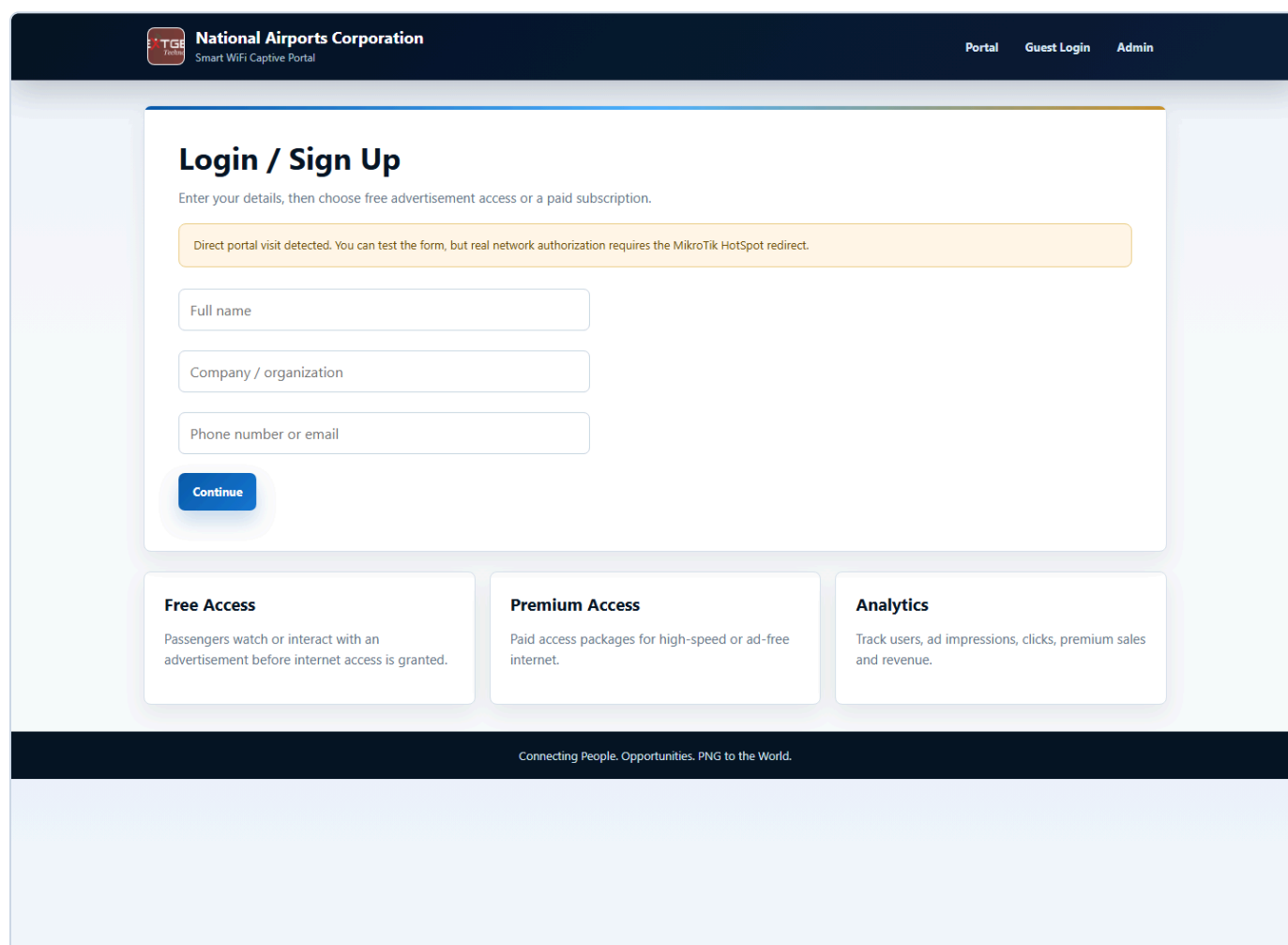
Figure 1: Open Portal page shown to users when they first access the WiFi service.

5. Open Portal and Guest Login

Open Portal This is the first page a user sees. It explains that the WiFi portal supports MikroTik HotSpot, UniFi external portal, and backend settings. It includes the login/sign up action and a link for administrator access.

Guest Login The guest form collects basic visitor information. This is important for accountability, usage reports, audit logs, and future customer service follow up.

Field	Why It Matters
Full name	Identifies the user or visitor.
Organization	Optional field for company, agency, airline, contractor, or visitor group.
Phone or email	Contact reference for audit, support, and business reporting.
Device/MAC and IP	Used to match the portal record to the device on the network when the controller provides it.



National Airports Corporation
Smart WiFi Captive Portal

Portal Guest Login Admin

Login / Sign Up

Enter your details, then choose free advertisement access or a paid subscription.

Direct portal visit detected. You can test the form, but real network authorization requires the MikroTik HotSpot redirect.

Full name

Company / organization

Phone number or email

Continue

Free Access

Passengers watch or interact with an advertisement before internet access is granted.

Premium Access

Paid access packages for high-speed or ad-free internet.

Analytics

Track users, ad impressions, clicks, premium sales and revenue.

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Figure 2: Guest login page where users enter their details before selecting free or paid access.

6. Admin Dashboard

The Dashboard is the main management screen. It summarizes users/sessions, advert impressions, advert clicks, and revenue. It also includes direct buttons for common tasks such as Settings, System Control, Ads, Plans, Portal Preview, and Setup Check.

The latest sessions table shows recent usage and updates live. It is useful for the airport operations team to see whether users are connecting and whether access is active or expired.

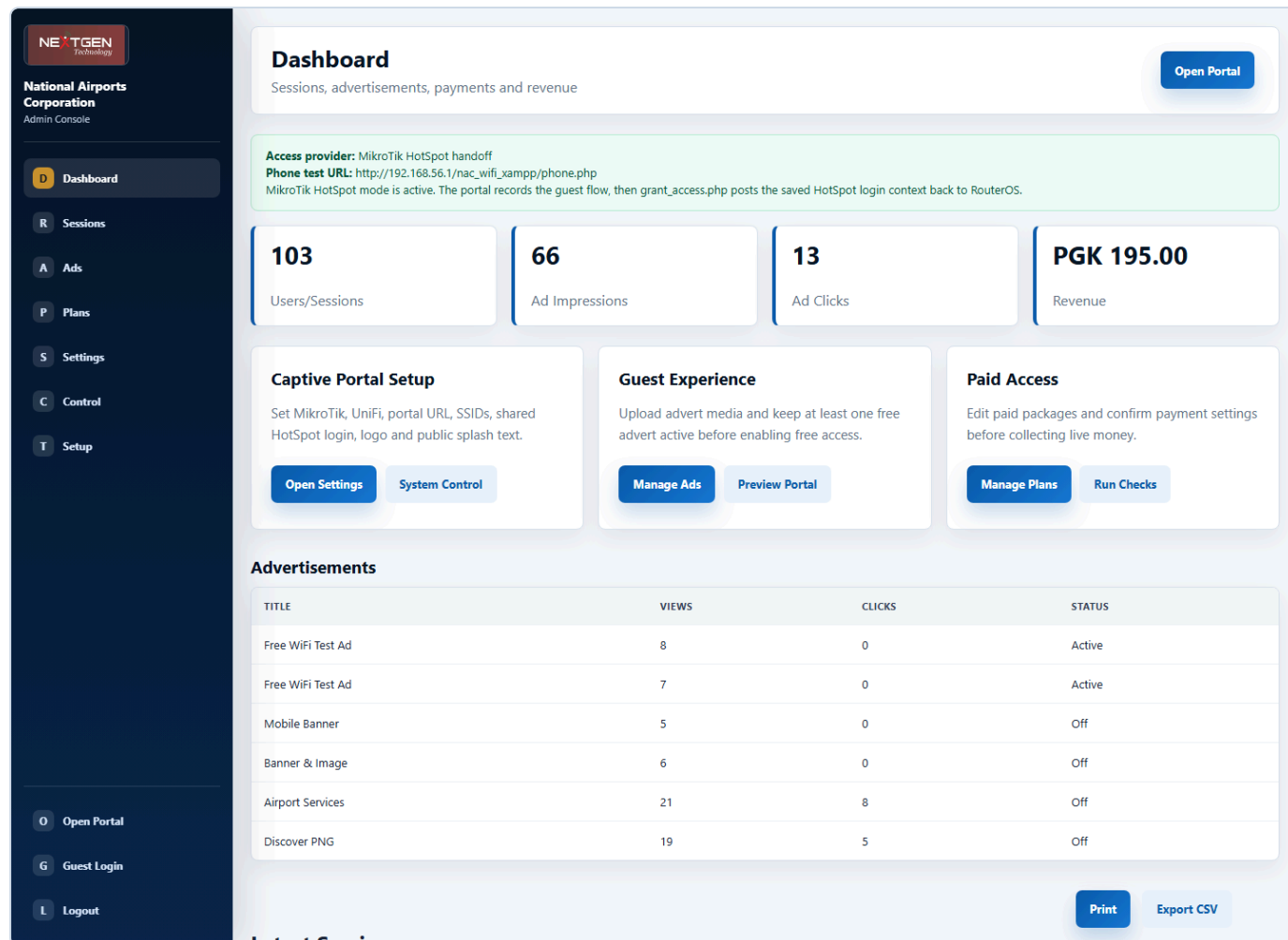


Figure 3: Dashboard with metrics, setup shortcuts, advertisements, and latest sessions.

Dashboard Controls

Control	Use
Print	Prints the current session view for management or manual filing.
Export CSV	Downloads session records for Excel or reporting.
Manage Sessions	Opens the Sessions Controller where sessions can be reviewed and deleted.

7. Sessions Controller

The Sessions Controller is the management page for internet access records. It is intentionally separate from the live dashboard so that delete actions are controlled and confirmed.

Session Controller
Print, export, review and delete captive portal sessions

Session Actions

Print Sessions Export CSV Delete Expired Delete All

Export includes the latest 1000 sessions. This page shows the latest 200 sessions.

Latest Sessions

GUEST	CONTACT	DEVICE/MAC	IP	TYPE	PACKAGE	STATUS	STARTED	EXPIRES	REMAINING	ACTION
Knight Wolf	710000000	MAC unavailable	192.168.56.1	free	Free Ad Access	expired	2026-05-20 09:28:48	2026-05-20 09:58:48	expired	Delete
Knight Wolf	710000000	MAC unavailable	192.168.56.1	free	Free Ad Access	expired	2026-05-20 09:17:24	2026-05-20 09:47:24	expired	Delete
Knight Wolf	710000000	MAC unavailable	::1	free	Free Ad Access	expired	2026-05-19 11:30:59	2026-05-19 12:00:59	expired	Delete
Spiderman	100100	MAC unavailable	192.168.56.1	free	Free Ad Access	expired	2026-05-19 10:33:15	2026-05-19 11:03:15	expired	Delete
Knight Wolf	710000000	MAC unavailable	192.168.56.1	free	Free Ad Access	expired	2026-05-19 09:20:11	2026-05-19 09:50:11	expired	Delete
Knight Wolf	710000000	MAC unavailable	192.168.56.1	free	Free Ad Access	expired	2026-05-19 08:25:51	2026-05-19 08:55:51	expired	Delete
Knight Wolf	710000000	MAC unavailable	192.168.56.1	free	Free Ad Access	expired	2026-05-19 08:00:23	2026-05-19 08:30:23	expired	Delete
Knight	710000000	MAC unavailable	192.168.56.1	free	Free Ad	expired	2026-	2026-	expired	

Figure 4: Session Controller for printing, exporting, and deleting session records.

Action	Explanation
Print Sessions	Prints a clean report view without the sidebar and buttons.
Export CSV	Exports the latest 1000 sessions for spreadsheet reporting.
Delete Expired	Removes old expired sessions and related access/payment records.
Delete All	Clears all session history. This should only be used after backup or during a reset.
Delete One	Removes one incorrect or test session.

Management note: Do not delete live operational records unless the data has been exported or backed up.

8. Advertisements and Free Access

Free access is controlled through active advertisements. When the user chooses free access, the system selects one active advertisement and shows it for a short countdown. After the countdown, the system records the advert view and grants free access for the configured time.

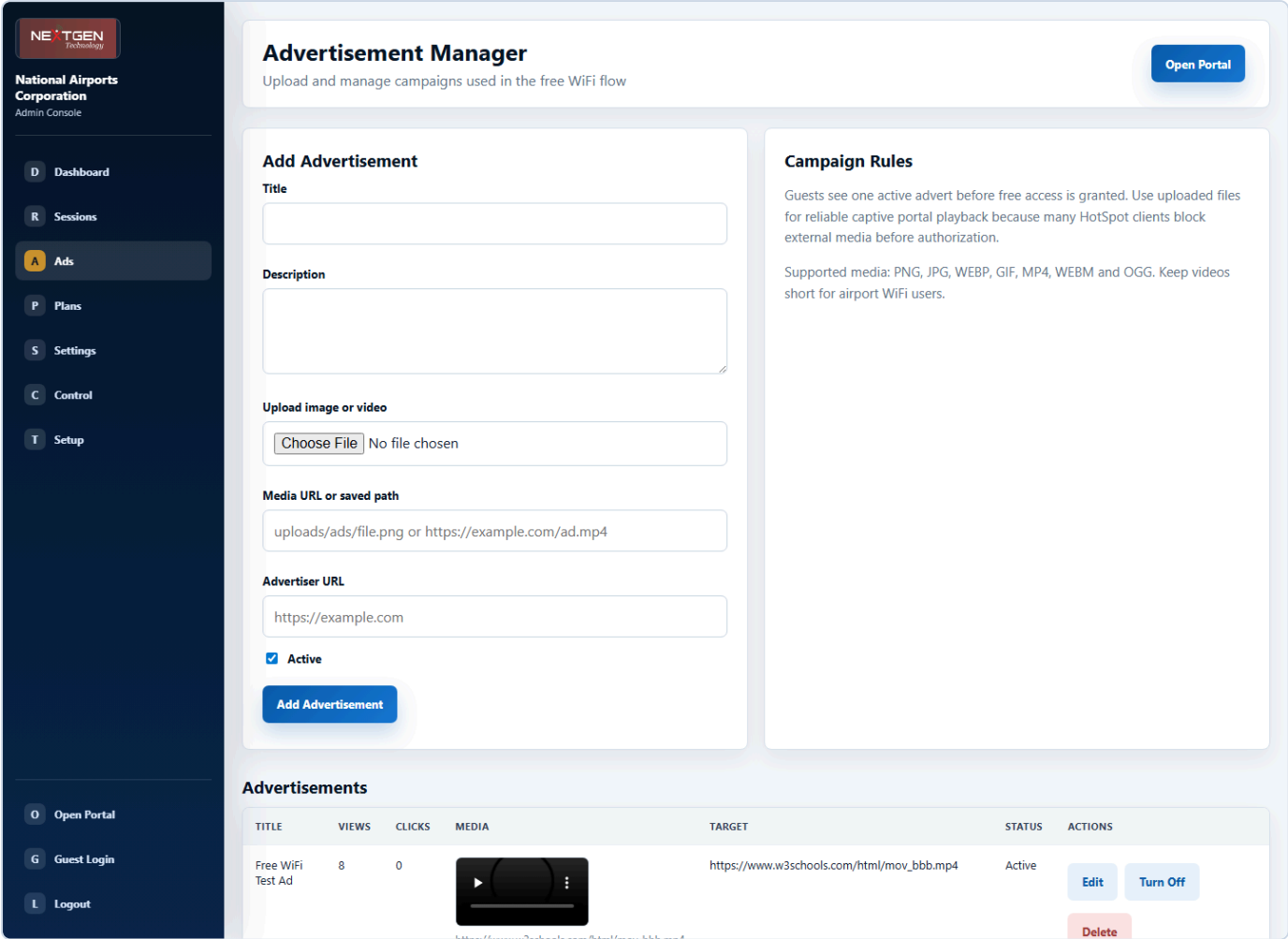


Figure 5: Advertisement Manager for upload, edit, activate, deactivate, and delete.

How Free Ads Work

- 1. Administrator uploads an image or video advert.
- 2. Administrator marks the advert active.
- 3. Guest selects Free Access.
- 4. Portal displays the advert and countdown.
- 5. System records the view and click if the user opens the advertiser link.
- 6. Portal creates an access grant and hands the user back to the router/controller.

For production captive portal use, uploaded local media is recommended because unauthenticated WiFi clients may not be allowed to load external images or videos before internet access is granted.

9. Plans, Paid Access, and Payments

Plans define paid WiFi packages such as 1 Hour, 3 Hours, or 1 Day. Each plan has a duration, price, active/off status, and sort order. The user sees only active plans on the paid access page.

Subscription Plans
Paid WiFi packages shown to guests

[Open Portal](#)

Add Plan

Plan name
2 Hours

Duration in minutes

Amount PGK

Sort order
0

☒ Active

[Add Plan](#)

Payment Readiness

Plans are shown on the paid access page. Keep at least one active plan before enabling premium access in a live captive portal.

For production payments, keep access granting tied to verified payment callbacks or manual admin approval.

Current Plans

NAME	MINUTES	AMOUNT	STATUS	ACTIONS
1 Hour	60	PGK 5.00	Active	Edit Turn Off Delete
3 Hours	180	PGK 10.00	Active	Edit Turn Off Delete
1 Day	1440	PGK 20.00	Active	Edit Turn Off Delete

Figure 6: Plans page where paid access packages can be edited, activated, or deleted.

How Paid Access Works

1. Administrator creates at least one active plan.
2. Guest selects Paid Subscription.
3. Guest chooses a package.
4. Payment is processed through configured provider or manual approval during testing.
5. After approval, the system creates a premium access grant for the selected duration.

Production payment rule: Live paid access should only be granted after verified payment callback, webhook, or approved manual payment confirmation.

10. Settings

The Settings page is the main configuration center. It is designed so an administrator can update portal branding, logo/profile image, admin username/password, deployment mode, controller address, public portal URL, SSIDs, MikroTik credentials, UniFi details, Spotipo details, and access instructions.

System Settings
Backend configuration for MikroTik, UniFi, Spotipo and production deployment

The default admin password is still active. Set ADMIN_PASSWORD_HASH or a strong ADMIN_PASSWORD in config.local.php before production use.

Portal Branding
Portal logo / profile image

Choose File No file chosen

Upload a square logo or profile image. PNG, JPG, WEBP and GIF are allowed.

Company name
National Airports Corporation

Portal title
Welcome to NAC WiFi

Portal subtitle
Smart WiFi Captive Portal

Footer text
Connecting People. Opportunities. PNG to the World.

Access instructions shown after approval
Your access has been approved. If this page was opened through a MikroTik HotSpot redirect, the router will now authorize this device.

Deployment Mode
Access provider
MikroTik HotSpot handoff

Controller mode
MikroTik RouterOS HotSpot

Controller admin URL
http://192.168.56.254

Controller enforcement status
Not integrated

Public portal host/IP
192.168.56.1

Portal path
/nac_wifi_xampp

Figure 7: Settings page for portal branding, deployment mode, admin account, MikroTik, UniFi, and Spotipo values.

Setting Area	What To Configure
Portal Branding	Company name, title, subtitle, footer text, and logo/profile image.
Admin Account	Admin username and secure password. The password is stored as a hash.
Deployment Mode	Select MikroTik HotSpot, UniFi external portal, Spotipo/UniFi API, or local demo.
Public Portal Host/IP	The IP or hostname that WiFi users can reach, for example <code>192.168.56.1</code> .
Portal Path	The folder path, normally <code>/nac_wifi_xampp</code> .
MikroTik HotSpot	Shared HotSpot username/password used for final RouterOS login handoff.
UniFi / Spotipo	Controller URL, API key, site ID, API base URL, and guest access durations.

11. System Control

System Control gives administrators the deployment links and the controlled HotSpot bridge file editor. This is not a general PHP file editor. It is intentionally limited to the network bridge file needed for MikroTik HotSpot deployment.

System Control
Controlled production tools for captive portal deployment

Deployment Links

Portal base:
`http://192.168.56.1/nac_wifi_xampp`

MikroTik entry URL:
`http://192.168.56.1/nac_wifi_xampp/mikrotik_entry.php`

Admin:
`http://192.168.56.1/nac_wifi_xampp/admin.php`

Setup check:
`http://192.168.56.1/nac_wifi_xampp/setup-check.php`

[Open Portal](#) [Run Setup Check](#) [Settings](#)

Controller Status

Enforcement status
Not integrated

[Save Status](#)

Mark integrated only after a real MikroTik or UniFi client completes access and appears authorized on the controller.

MikroTik HotSpot Bridge File

Upload this file as RouterOS HotSpot `hotspot/login.html`. It passes RouterOS variables into this PHP portal, then the portal completes the guest/ad/payment flow.

```
<!doctype html>
<html>
<head>
  <meta charset="utf-8">
  <meta name="viewport" content="width=device-width, initial-scale=1">
```

[Save Bridge File](#)

[Regenerate From Current Portal URL](#)

Figure 8: System Control page with deployment links, enforcement status, and MikroTik bridge file editor.

Key Controls

Control	Use
Portal Base URL	The address used by APs/controllers to send users to the portal.
MikroTik Entry URL	The URL used by RouterOS HotSpot login.html to pass HotSpot variables to PHP.
Enforcement Status	Internal readiness flag showing whether router/controller enforcement has been proven.
Regenerate Bridge File	Builds hotspot/login.html using the current portal URL.

12. MikroTik Captive Portal Integration

MikroTik RouterOS HotSpot is the current recommended direct integration path. RouterOS controls internet access. The PHP portal handles branding, guest registration, ads, payment flow, and session records.

Required MikroTik Items

- MikroTik router or CHR with HotSpot configured.
- WiFi AP or bridge connected to the HotSpot network.
- A shared HotSpot user that the portal can post back to RouterOS after approval.
- The portal server IP reachable by HotSpot clients.
- The generated `hotspot/login.html` uploaded to RouterOS HotSpot files.

MikroTik Deployment Steps

1. Configure the HotSpot network on MikroTik.
2. Create the shared HotSpot username/password.
3. In Admin Settings, select `MikroTik HotSpot handoff`.
4. Set Public Portal Host/IP and Portal Path.
5. Enter the shared HotSpot username/password.
6. Open System Control and regenerate the HotSpot bridge file.
7. Upload `hotspot/login.html` to the MikroTik HotSpot file area.
8. Connect a test client to WiFi and open `http://neverssl.com`.
9. Complete guest login, free advert, or paid flow.
10. Confirm the client appears in MikroTik active HotSpot users and has internet access.

Result: The user is not just shown a success page. RouterOS is used to authorize real network access.

13. UniFi Captive Portal Integration

UniFi integration depends on the UniFi Network version, gateway/AP model, and available external portal/API method. The system is ready to hold UniFi controller settings and route the guest flow through the portal. Final authorization must be connected to the confirmed UniFi method when hardware arrives.

Required UniFi Items

- UniFi access point or UniFi gateway/access point setup.
- UniFi Network application/controller reachable from the portal server.
- Guest WiFi network with captive portal or external portal enabled.
- External portal URL pointing to the NAC portal.
- UniFi site ID and API access method confirmed for the installed version.

UniFi Deployment Steps

1. Adopt the UniFi AP into the controller.
2. Create the guest WiFi SSID.
3. Enable guest/captive portal or external portal mode.
4. Set the external portal URL to the NAC portal base URL.
5. In Admin Settings, choose UniFi external portal or Spotipo/UniFi depending on the final controller method.
6. Enter controller URL, API key or token, and site ID.
7. Connect a test client and confirm the redirect opens the NAC portal.
8. Complete free or paid access and confirm the client is authorized by UniFi.

Important: UniFi final authorization must be tested with the real purchased hardware and controller version. Do not mark enforcement as integrated until a real WiFi client receives internet through UniFi after portal approval.

14. Setup Check and Production Readiness

The Setup Check page verifies the database, required files, controller values, payment status, API readiness, and required tables. It is used before live testing and after any deployment move.

Setup Check
Deployment readiness and configuration status

[Open Portal](#)

System Readiness
Use this page after copying the project to XAMPP and before switching live MikroTik, UniFi, or Spotipo authorization on.

[Open Portal](#) [Admin Dashboard](#) [Settings](#)

Checks

CHECK	STATUS	DETAILS
Database	OK	Connected successfully.
Application folder	INFO	C:\Users\akalisik\xampp\htdocs\nac_wifi_xampp
Environment	INFO	local
Access provider	INFO	MikroTik HotSpot handoff
Phone entry URL	INFO	http://192.168.56.1/nac_wifi_xampp/phone.php
Admin URL	INFO	http://192.168.56.1/nac_wifi_xampp/admin.php
Admin password	WARN	Default password is still active. Change config.local.php before production.
Controller admin URL	INFO	http://192.168.56.254
Controller enforcement	WARN	Portal tracking is working, but router allow/block enforcement is not integrated yet.
MikroTik HotSpot login	OK	Shared HotSpot login is available.
UniFi direct controller	WARN	Not configured. Add UniFi controller URL and API key after UniFi quotation/procurement.
Spotipo API base	INFO	https://api.spotipo.com
Spotipo site ID	WARN	Missing.

Figure 9: Setup Check page for deployment readiness and system health.

Production Requirement	Action
Admin password	Change default admin password from Settings before production.
Portal URL	Set host/IP and path that clients can reach from guest WiFi.
Controller mode	Select MikroTik HotSpot or UniFi integration mode.
Advert media	Upload at least one active local advert for free access.
Plans	Create at least one active paid package if paid access is offered.
Payment	Use verified payment callback or manual approval process for live money.
Test client	Test from a real WiFi client behind the AP/router, not only from the server browser.
Controller proof	Confirm the client appears as authorized on MikroTik or UniFi.

15. Access Types Explained

Free Advertisement Access

Free access is used when NAC or a sponsor wants to provide limited WiFi in exchange for advert exposure. The user must watch or interact with an advert before access is granted. The system records advert views and clicks.

Paid Access

Paid access is used when users need longer or premium connectivity. The administrator controls the plan name, price, and duration. Payments should be verified before final access is granted.

Manual or Testing Access

During office testing, the system can record sessions without final router enforcement. This is useful for checking screens and reports, but production access must be tested through MikroTik or UniFi.

16. Roles and Responsibilities

Role	Responsibility
Managing Director / Management	Review service model, approve deployment, monitor reports and revenue/ad performance.
IT Administrator	Configure MikroTik/UniFi, portal settings, passwords, HotSpot bridge file, backups, and security.
Marketing / Commercial Team	Upload adverts, activate campaigns, monitor views and clicks.
Finance / Revenue Team	Review paid package records, payment process, and exported CSV reports.
Support Staff	Assist guests and check whether sessions are active or expired.

17. Final Recommendation

The system is ready as a captive portal application layer and administrator console. The remaining production proof depends on the arrival and configuration of MikroTik or UniFi hardware. Once the AP/router is installed, the most important test is a real client test from the guest WiFi network.

Go-live rule: Do not consider the system live until a phone or laptop connects to the guest WiFi, is redirected to the NAC portal, completes the flow, appears authorized on the router/controller, and receives internet access.

Management Summary

This portal gives NAC a controlled, branded, reportable WiFi access system. It supports free sponsored advertising access, paid package access, administrator reporting, and a path to MikroTik or UniFi captive portal enforcement. The system is structured so non-technical staff can manage day-to-day content while IT controls the router/controller integration.